A Robust Model for MODIS and Landsat Image Fusion Considering Input Noise

Zhenyu Tan, Meiling Gao, Jun Yuan, Liangcun Jiang, and Hongtao Duan

***Abstract*— Significant progress has been made in spatiotemporal fusion for remote sensing images; however, most models require inputs to be free of clouds and without missing data, considerably confining their applications in practice. Due to recent advances in deep learning technologies, powerful modeling capabilities could be leveraged to bring potential solutions to this problem. This article proposes a novel architecture named the robust spatiotemporal fusion network (RSFN) based on the generative adversarial network and attention mechanism with dual temporal references to automatically handle input noise. The RSFN only needs one coarse-resolution image on the prediction date and two referential fine-resolution images before and after the prediction date as model inputs. Most notably, there is no special restriction attached on the data quality of referential images. The comparison with other models demonstrates the effectiveness of the RSFN model quantitatively and visually in four study areas using MODIS and Landsat images. Two main conclusions can draw from the experiments. First, the input data noise hardly affects the prediction results of the RSFN, and the RSFN can gain a comparable or even higher accuracy; conversely, the other methods only show limited resistance to input noise. Second, the RSFN with cloud-contaminated references outperforms the other models with cloud-free references after data filtering in the same study area during the same period. The satellite data quality usually varies significantly; the model robustness and fault tolerance are considered critical for actual applications. The RSFN is a simple end-to-end deep model with high accuracy and fault tolerance designed for spatiotemporal fusion with imperfect data inputs, showing promising prospects in practical applications.**

***Index Terms*— Data fusion, deep learning, noisy data, robust spatiotemporal fusion network (RSFN), robustness, spatiotemporal.**

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I. Introduction

ADVANCES in sensor technology render the spatial and temporal resolutions of remotely sensed images much higher than before; however, acquiring high spatial and high temporal resolution images simultaneously is still a major challenge for the remote sensing community [1]–[3]. Satellite sensors with high spatial resolution usually have a low temporal resolution; in contrast, a high temporal resolution is usually accompanied by a low spatial resolution, such as Landsat and MODIS data [4], [5]. Although there will be more satellites with high spatiotemporal resolution in the future, satellite data mining with long-term historical archives is also of great significance. Under this circumstance, spatiotemporal fusion models (STFMs) partially fill this gap by combining the ground details from sensors with higher spatial resolution and ground changes from sensors with a higher temporal resolution to synthesize high spatiotemporal image series. Since the spatial resolution gap between two data sources is often quite large, usually, at least one fine-resolution image temporally close to the prediction date is needed to provide ground details as a reference. Then, with the fine-resolution reference, the coarse-resolution image on the prediction date, and, sometimes, other auxiliary data, the fine-resolution image on the prediction date can be derived [1], [6], [7]. The synthesized fine-resolution images can serve fine-grained resource and environment monitoring applications, such as precision agriculture, land cover classification, and ecological studies, simultaneously incorporating both the surface details and temporal variations [1].

Numerous STFMs have been proposed over the past decades to enhance the spatial and temporal resolutions of existing remote sensing image archives. They can be roughly classified into five categories [6], [7]: 1) transformation-based methods, such as the wavelet transform [8]; 2) weight-function-based methods, such as the spatial and temporal adaptive reflectance fusion model (STARFM) [9], the spatial–temporal adaptive algorithm for mapping reflectance change (STAARCH) [10], and the enhanced STARFM (ESTARFM) [11]; 3) unmixing-based methods, such as the flexible spatiotemporal data fusion (FSDAF) method [12] and the linear spectral unmixing-based spatiotemporal data fusion model (LSUSDFM) [13]; 4) Bayesian-based methods [14], [15]; and 5) learning-based methods [7], [16].

The first four categories, which were designed and developed based on rigorous geological processes under some assumptions [6], are usually regarded as the conventional

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methods. Those models tend to become unreliable once the hypothetical assumptions are not satisfied. In contrast, learning-based methods are instantiated with the learned rules from massive actual training samples. They are independent of the ideal assumptions and usually should be practically more resistant to noise.

Currently, most of the STFMs require clean noiseless inputs, and this is understandable because the conventional models are mostly established at the pixel level, including defective pixel values could complicate the fusion problem itself. Therefore, the ideal inputs are regarded as the basic prerequisite to simplify the modeling process, and the input noise, or cloud contamination most of the time, is barely considered in model development. However, optical images are highly susceptible to clouds and rain [17]. The simplest solution for this problem is just to remove the noisy images from the collected dataset to prepare ready-to-use model inputs [18]. However, after data filtering, the acquisition date of the fine-resolution reference is temporally far from the prediction date [19]. During such a long period, sometimes, the ground features have changed significantly, and the information from the referential image might be inaccurate. Consequently, the prediction accuracy is impaired. Another feasible but a little complicated solution is to perform data repair before fusion, such as removing unscanned gaps and cloud-contaminated pixels, to reconstruct the noiseless inputs. For example, the satellite data integration (STAIR) model uses linear regression and adaptive-average correction to impute the missing pixel values on multitemporal Landsat images before the regular fusion [20]. For one thing, involving the data repair process in advance can complicate the whole fusion process; for another, errors of data repair can directly affect the following fusion accuracy. Hence, it is necessary to develop fusion models that can automatically handle the input noise and improve the model robustness in an end-to-end manner.

Nowadays, learning-based methods receive much attention in the geoscience community due to their powerful modeling capability [21], [22]. Unlike conventional methods using handcrafted rules, learning-based methods can automatically establish an implicit relationship between input(s) and output(s) from archived datasets with the inherent learning mechanism. Sparse representation-based methods, such as the sparse representation-based spatiotemporal reflectance fusion model (SPSTFM) [23], learn two sets of dictionaries for fine and coarse image pairs and then utilize the shared sparse coefficient and the learned dictionary for fine-resolution images to rebuild the prediction. Machine learning-based models employ advanced machine learning techniques to merge relevant information and derive the expected fine-resolution images. Examples include the spatiotemporal fusion using a deep convolutional neural network (STFDCNN) [24], the two-stream convolutional neural network (CNN) for spatiotemporal image fusion (StfNet) [25], the enhanced deep convolutional STFM (EDCSTFN) [16], the spatiotemporal fusion via CycleGAN-based image generation (CycleGAN-STF) [26], and the GAN-based STFM (GAN-STFM) [27]. Beyond the aforementioned spatiotemporal fusion, deep learning is also extensively employed for other types of remote sensing image fusion, such as pan-sharpening [28], spatial resolution enhancement among different bands of one sensor [29], and super-resolution between images of different satellites [30]. In a word, the CNNs and generative adversarial networks (GANs) are shown as effective modeling tools for remote sensing image fusion and reconstruction problems [31]–[33].

This work of STFMs mainly focuses on improving model accuracy and enhancing image quality; little attention is paid to input noise and model robustness. Thus, this article proposes a novel model for remote sensing images named the robust spatiotemporal fusion network (RSFN) to automatically filter input noise from dual temporal referential images and prevent predictions from being corrupted. Compared with recent studies, the main contributions of this article are summarized as follows.

1) The CNN and GAN techniques and the attention mechanism are leveraged to build the RSFN with high fault tolerance. To the best of our knowledge, the RSFN is the first end-to-end deep STFM that can automatically handle input data noise without impairing prediction accuracy.
2) The importance of timeliness of referential images in the spatiotemporal fusion was confirmed by the fact that the RSFN with cloud-contaminated references outperforms the other models with cloud-free references after data filtering in the same study area during the same period. Since there is no special restriction attached on the data quality of referential images, the RSFN model can ensure the timeliness of input references as much as possible.
3) The RSFN model can be employed in cloud-prone regions with fairly high accuracy compared with the existing models, significantly expanding the application scope of spatiotemporal fusion.

The remainder of this article is organized as follows. The basics of the employed deep learning techniques and the detailed architecture of the proposed RSFN are introduced in Section II. Section III presents the details of the experiments and results. The relevant discussion is presented in Section IV. The conclusions and future work are summarized in Section V.

II. Materials and Methods

A. CNNs and GANs

CNNs, as a major class of deep learning models, have dominated the computer vision domain in various image-related tasks and have proven their superiority [34]. A typical CNN model is composed of several convolution building blocks, and each block usually includes a convolution layer and an activation layer. CNN models usually employ a mass of convolution filters to extract self-adaptive features for a specific task. Higher level features can be automatically extracted with deeper layers; eventually, the CNN model can converge to an optimal solution of the task [35]. The parameters of those filters in a deep model are optimized in the training process with the backpropagation algorithm. The activation layer in each block is used to introduce nonlinearity, pushing the

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network nodes to learn more complex features of the data. The commonly used activation function in a CNN is the rectified linear unit (ReLU) and its variants [36].

The combination of GANs and CNNs is often used for image generation and reconstruction tasks. A classic GAN model consists of a generator and a discriminator [37]. The core idea of GANs is the competition between the two subnets in the form of a zero-sum game, where one's gains are against another's losses. The training process of the generator is dynamically propelled through the training of the discriminator. Eventually, the generator can generate the resemblance with similar statistical distribution as the training samples, and the discriminator cannot easily distinguish the generated fake data and the actual training samples. The original GAN suffers from poor stability and mode collapse [38]; hence, many variants of GANs from different perspectives have been proposed to improve the performance of the vanilla GAN. For example, the conditional GAN (CGAN) [39] introduces condition labels to both the generator and discriminator to guide the direction of generated data. The least-squares GAN (LSGAN) [40] proposes a new loss function to replace the standard min–max loss to enhance the GAN's stability. The loss function of the conditional LSGAN is formulated as follows:

$$\begin{aligned}\min_D V_{\mathrm{LSGAN}}(D) &= \frac{1}{2}\mathbb{E}_{\boldsymbol{x}\sim p_{\mathrm{data}}(x)}\left[(D(\boldsymbol{x}\mid \boldsymbol{y})-b)^2\right] \\ &\quad +\frac{1}{2}\mathbb{E}_{\boldsymbol{z}\sim p_z(z)}\left[(D(G(\boldsymbol{z}\mid \boldsymbol{y}))-a)^2\right] \\ \min_G V_{\mathrm{LSGAN}}(G) &= \frac{1}{2}\mathbb{E}_{\boldsymbol{z}\sim p_z(z)}\left[(D(G(\boldsymbol{z}\mid \boldsymbol{y}))-c)^2\right] \end{aligned} \tag{1}$$

where G and D denote the generator and discriminator, respectively. Tensors $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$ denote the ground truth, conditional label, and noisy input, respectively. Constants a, b, and c satisfy the conditions of $b-c=1$ and $b-a=2$. Moreover, spectral normalization [41] is introduced in GANs to stabilize the training process by enforcing a Lipschitz constraint. This article integrates the techniques of the CGAN, LSGAN, and spectral normalization into the proposed RSFN to generate realistic remote sensing images.

B. Attention Mechanism

Inspired by the human vision system, the concept of attention broadly refers to a mechanism by which a deep network can weigh extracted features on the grounds of the levels of importance in a specific task [42], [43]. Attention mechanisms can be used in different kinds of deep models for different purposes. An attention mechanism contributes to a computer vision problem by focusing on the key information of an image instead of giving every pixel the same attention. A CNN can be used to derive attention weights for both image classification and generation. This article utilizes the attention mechanism to guide the fusion model to automatically filter out input data noise and synthesize noiseless fine-resolution feature maps at the feature level, thus improving the fault tolerance and robustness of the model.

How can an attention mechanism be employed in an STFM? Taking Landsat and MODIS image fusion as an example, if there are Landsat images available at time t_0 and t_2, and MODIS data available at times t_0, t_1, and t_2, the STFM aims to generate a Landsat-like image at prediction time t_1. Since the input images might be contaminated with clouds or have missing pixel values, the two references at the preceding and succeeding time of t_1 can be weighted based on an attention mechanism to synthesize a clean uncorrupted referential image. To address noisy inputs and ground temporal changes, two types of information can be input into a deep network to derive the corresponding weights. One is the Landsat or MODIS data themselves, and the other is the implicit relationship between Landsat and MODIS images at referential and prediction time. In other words, first, the attention subnet should have the ability to identify the noisy pixels in the referential images. Second, similarities between observations at prediction time t_1 and referential time t_i ($i \neq 1$) can be utilized to drive the attention weights. Theoretically, the attention subnet should derive weights of zero for the areas with noise, small weights for the areas where significant ground changes are observed between the Landsat reference and MODIS on the prediction date, and large weights for the areas where the Landsat reference and MODIS look similar. Based on the above consideration, the Landsat reference at t_i and the MODIS image at t_1 are stacked and delivered to the attention subnet to derive the corresponding weights at time t_i. Since two phases of Landsat-MODIS data pairs are used here, noiseless clean ground features can be generated using the corresponding weights at t_0 and t_2 under the assumption that the noisy pixels in the two referential images appear at the same locations, which is a small probability event on a large scale.

C. RSFN Architecture

The RSFN is a CGAN model essentially, so it is likewise composed of a generator and a discriminator, as shown in Fig. 1. The generator aims to synthesize the Landsat-like image at prediction time t_1, while the discriminator seeks to differentiate between the actual Landsat observation and the predicted Landsat-like image. The discriminator is a typical binary classification network composed of several convolutional residual blocks. The inputs of the discriminator are either the stacked images of the Landsat-like prediction and MODIS at t_1 or the stacked actual Landsat and MODIS pairs at t_1. The MODIS image here serves as the condition label. The expected outputs of the discriminator are either zero or one, indicating the false and true status of the fine-resolution inputs, respectively. The residual building block employed in the discriminator can mitigate the vanishing gradient and degradation problems of a deep network compared with the plain convolution layer [44]. The residual block designed for discriminator (DResBlock) is illustrated in Fig. 2(a). Conv ($kisj$) represents the convolution layer, where the hyperparameter ki in parentheses denotes the $i \times i$ convolution kernel, and sj means that the convolution stride is set to j. MaxPool ($k2$) represents the max-pooling operator with a kernel size of 2. Max pooling aims to shrink the spatial dimension of feature maps, and thus, it reduces computational consumption. The batch normalization used before each leaky

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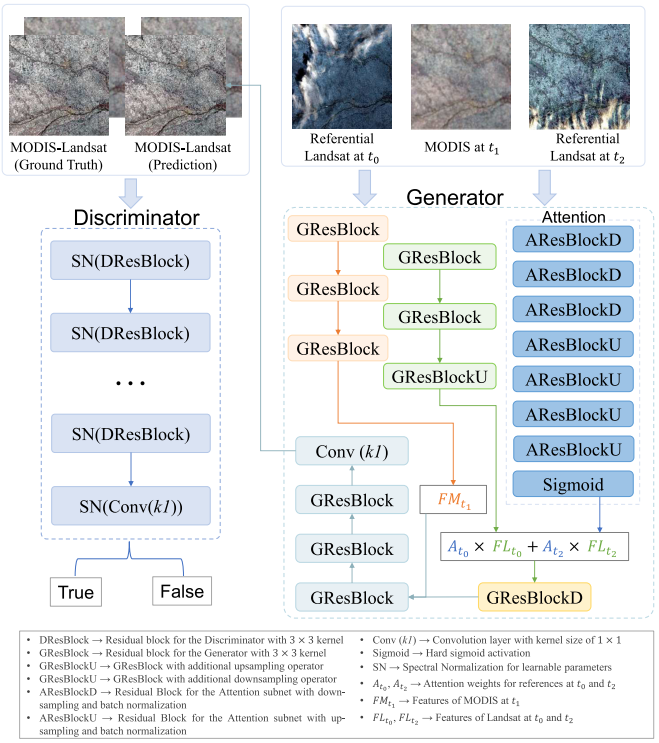


Fig. 1. Architecture of the RSFN model. (The RSFN model is composed of a generator and a discriminator, and the attention subnet is part of the generator. GResBlock is designed for the generator, and DResBlock is designed for the discriminator. Conv ($k1$) represents the convolution layer with 1×1 kernel size. BN and SN denote the batch normalization and spectral normalization, respectively. A_{t0} and A_{t1} represent the output weights of the attention subnet corresponding to the referential time of t_0 and t_1, respectively. FM_{t_i} and FL_{t_i} denote the feature maps corresponding to the MODIS and Landsat images at time t_i.)

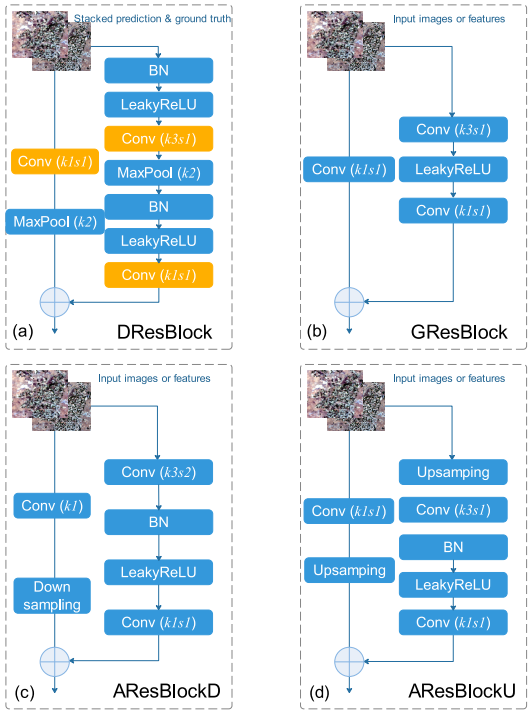


Fig. 2. Residual building blocks for discriminator [(a) DResBlock], generator [(b) GResBlock] and attention subnet [(c) AResBlockD and (d) AResBlockU for downsampling and upsampling, respectively]. (Conv ($kisj$) represents the convolution operator, where the hyperparameter ki in parentheses denotes the $i \times i$ convolution kernel, and sj means that the convolution stride is set to j. MaxPool ($k2$) represents the max-pooling operator with a kernel size of 2. BN denotes the batch normalization for the output features of the previous layer. The weights of the orange convolution layers are normalized with spectral normalization to stabilize the GAN training.)

ReLU activation is intended to generalize the model. Besides, spectral normalization is utilized to stabilize the training of the discriminator. The 1×1 convolution layer at the end of each residual block can decrease the number of model's parameters, thus reducing the model complexity and reducing overfitting to a certain degree.

The generator adopts the classic "encoder-fusion-decoder" pattern, as illustrated in Fig. 1. The inputs include one MODIS image on the prediction date and two Landsat referential images acquired before and after the prediction date. Two things need to be noticed: first, not like other models, the RSFN only needs one MODIS image for model inputs, and data quality requirements are only applied to the MODIS image; second, both images should be resampled to have the same size before entering the model. With the three inputs being prepared, first, the encoder subnets in green and orange extract the high-dimensional features from the input Landsat and MODIS images, respectively. Then, all the features are merged in the abstract feature space. Finally, the merged features are entered into the decoder subnet in blue to restore to the raw image pixels. The generator also comprises several convolutional residual blocks (GResBlocks), with an architecture slightly different from that of DResBlocks, as shown in Fig. 2(b). Normalization is not employed in the encoder and decoder subnets because batch normalization can forcibly change the distribution of the original data, which can certainly prejudice the final image reconstruction. Notably, the last residual block (GResBlockU) for the Landsat image encoder in green employs an upsampling operator to expand the feature size twice. Then, after weighting the Landsat references at the subpixel scale, the generated noiseless referential features are tweaked and downsampled with another residual block (GResBlockD), as shown in yellow in Fig. 1.

The attention subnet plays a significant role in the generator to derive corresponding weights for the two Landsat references and expects to yield clean uncorrupted referential feature maps in the fusion process. The inputs of the attention subnet are the stacked Landsat image at t_i ($i \neq 1$) and the MODIS image at t_1. It is expected to not only identify the noisy pixels in the Landsat but also qualify the similarity between MODIS and Landsat by entering both Landsat and MODIS images to the attention subnet. The building block used in the attention subnet is similar to GResBlock, but batch normalization is added to generalize the weight calculation. Moreover, the classical "hourglass" architecture is employed here, to wit: there are upsampling and downsampling operators involved in the attention subnet. This scaling in the spatial dimension not only can weaken the boundary effect of the input noisy regions, to a certain degree, but also can reduce memory consumption.

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The outputs of the attention subnet are the normalized weights corresponding to the two references ranging in (0, 1) with the help of sigmoid activation, denoted with A_{t_0} and A_{t_2}. Next, the clean and uncorrupted referential features are synthesized with the attention weights and the feature maps of Landsat references extracted from the generator encoder using the following equation:

$$\mathrm{FL}_{t_{02}} = A_{t_0} \times \mathrm{FL}_{t_0} + A_{t_2} \times \mathrm{FL}_{t_2} \tag{2}$$

where FL_{t_i} denotes the Landsat feature maps at time t_i. All arithmetic operations are conducted elementwise. Finally, the weighted referential features $\mathrm{FL}_{t_{02}}$ and MODIS features FM_{t_1} at t_1 are stacked and input into the generator decoder to produce the ultimate prediction. It is worth mentioning that the assumption on the attention that noisy pixels in the two referential images rarely appear at the same locations may not always satisfied, but the GAN-based architecture has certain ability to conjecture the missing information in this case [45]. That is the reason why both the GAN architecture and attention mechanism were employed in the RSFN model.

D. Loss Function

A compound loss function, including image-based losses and GAN losses, is designed to train the RSFN model. The image-based losses further consist of content loss, feature loss, spectra loss, and vision loss, similar to the compound loss function employed in the GAN-STFM model [27]. The overall losses are shown in the following equation:

$$\begin{aligned} \min_G V(G) &= \mathcal{L}_{\mathrm{IMG}} + \lambda \mathcal{L}_{\mathrm{GAN}} \\ \mathcal{L}_{\mathrm{IMG}} &= \mathcal{L}_{\mathrm{Content}} + \alpha \mathcal{L}_{\mathrm{Feature}} + \beta \mathcal{L}_{\mathrm{Spectra}} + \gamma \mathcal{L}_{\mathrm{Vision}} \end{aligned} \tag{3}$$

where α, β, γ, and λ are the weighting coefficients. The content loss is calculated using the pixelwise mean square errors (mses) with the Sobel operator to preserve the spatial details, formulated as follows:

$$\mathcal{L}_{\mathrm{Content}} = \frac{1}{N} \sum_{i=1}^{N} \left[f_{\mathrm{Sobel}}(\boldsymbol{x}) - f_{\mathrm{Sobel}}(\boldsymbol{y}) \right]^2 \tag{4}$$

where $\boldsymbol{x}$ and $\boldsymbol{y}$ denote the prediction and ground-truth images, respectively; N represents the number of pixel values. The feature loss, also called the perceptual loss, is calculated in the abstract feature space because studies have shown that the feature-based loss function can preserve more high-level contextual and perceptual details [46]. The feature loss calculates the pixelwise mses of the feature maps extracted with a pretrained autoencoder. The autoencoder employed in this article also adopts the classical "hourglass" architecture, composed of an encoder and decoder with eight building blocks in total, which is the same as the one used in the GAN-STFM model [27]. The feature loss function is formulated as follows:

$$\mathcal{L}_{\mathrm{Feature}} = \frac{1}{N} \sum_{i=1}^{N} \left[f_{\mathrm{Encoder}}(\boldsymbol{x}) - f_{\mathrm{Encoder}}(\boldsymbol{y}) \right]^2 \tag{5}$$

where f_{Encoder} denotes the encoder subnet in the pretrained autoencoder network. The spectra loss function works by measuring the cosine similarity between the prediction and ground truth along the feature channel dimension, as shown in the following equation:

$$\mathcal{L}_{\mathrm{Spectra}} = \frac{1}{N} \sum_{i=1}^{N} \left[\arccos\left(\frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\| \|\boldsymbol{y}\|} \right) \right]. \tag{6}$$

From a certain point of view, the content and feature losses perform in horizontal space and maintain the ground texture similarity, while the spectra loss works in the vertical direction and encourages spectral similarity. The vision loss is borrowed from the computer vision domain to visually measure the image similarity. The vision loss can be calculated with the multiscale structural similarity (MS-SSIM) [47], as shown in the following equation:

$$\mathcal{L}_{\mathrm{Vision}} = \boldsymbol{I} - \left[l_M(\boldsymbol{x}, \boldsymbol{y}) \right]^{\alpha_M} \cdot \prod_{i=1}^{M} \left[c_i(\boldsymbol{x}, \boldsymbol{y}) \right]^{\beta_i} \left[s_i(\boldsymbol{x}, \boldsymbol{y}) \right]^{\gamma_i} \tag{7}$$

where I denotes the multidimensional tensor of ones. $l_i(x, y)$, $c_i(x, y)$, and $s_i(x, y)$ are the quantified luminance, contrast, and structure of predicted and expected images at the ith scale, respectively. M denotes the highest scale. α_i, β_i, and γ_i are the corresponding constant weight coefficients. Generally, the compound loss function in image generation tasks performs better than the pixel-based mse loss.

E. Datasets

Four datasets covering four study sites were used in the following experiments. The first two datasets of Landsat 8 OLI and MODIS (MOD09GA) surface reflectance images, covering Tianjin, China (117.05°E∼117.62°E,38.77°N∼39.28°N) and Nebraska, USA (97.63°W∼98.35 °W,41.62°N∼42.37°N), from 2015 to 2018, were collected for the experiments. Tianjin city is an example of a metropolitan region with various ground objects; Nebraska is an example of an agricultural region with distinct seasonal ground changes. Since the Landsat and MODIS surface reflectance products have undergone basic data preprocessing, only additional image cropping, reprojection, and registration are needed to make the products match in the same spatial reference and spectral bands. Both MODIS and Landsat images were resampled to a spatial resolution of 30 m with bilinear interpolation. Each image is composed of six spectral bands ranging from blue to infrared. The correspondence relationship between Landsat and MODIS image bands can be found in [48]. Next, all the images after preprocessing were filtered based on the condition that cloud coverage is less than 45% per scene, and then, the MODIS-Landsat pairs were ordered chronologically and grouped according to the temporal distance. Because the fusion model needs two references to perform fault-tolerant processing, images from three time points are required for model inputs, and certainly, the temporally closest three image pairs were grouped. Finally, there are 19 data groups with a size of 1892 × 1630 available in the Tianjin dataset. Of these groups, 13 image groups ranging from 2015 to 2017 were used for training, and the remaining six groups were used for validation. Validation data were not involved in the training process. The Nebraska dataset consists of 33 image groups

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TABLE I
NUMBER OF LEARNABLE PARAMETERS FOR EACH DEEP MODEL

StfNet	EDCSTFN	RSFN	
		Generator	Discriminator
118310	284134	284490	1244813

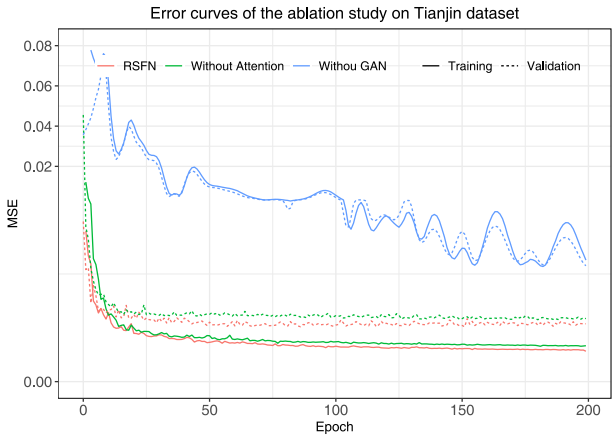


Fig. 3. Error curves of the ablation study in the Tianjin area.

with a size of 2752 × 1984. Similarly, 25 groups ranging from 2015 to 2017 were used for training, and the remaining eight groups were used for validation. The maximum and mean ratio of the noisy pixels (no data, cloud, shadow, and so on) are 27.45% and 9.10% of the Tianjin dataset and 40.00% and 8.32% of the Nebraska dataset. The cloud cover ratio in the Tianjin dataset is more evenly distributed than that in the Nebraska dataset. Thick clouds only occur in some groups of the Nebraska dataset over a large area.

Moreover, another two open-source datasets located in the Coleambally Irrigation Area (CIA) and Lower Gwydir Catchment (LGC) [49] of New South Wales, Australia, were also used in the experiments. The CIA dataset includes 17 cloud-free MODIS-Landsat image pairs from 2001 to 2002 with a size of 2040 × 1720, and the LGC comprises 14 MODIS-Landsat data pairs from 2004 to 2005 with a size of 2720 × 3200. Similarly, the spatial resolution was also resampled to 30 m, and each group includes two referential image pairs and one pair of images on the prediction date. After grouping all the datasets, there are 15 and 12 groups for the CIA and LGC datasets, respectively. For the CIA dataset, ten data groups of data acquired in 2002 were used for training, and the remaining five groups in 2001 were used for validation. For the LGC dataset, seven groups in 2004 were used for training, and the remaining five groups in 2004 and 2005 were used for validation.

III. EXPERIMENTS AND RESULTS

A. Experiments

1) Experiment Design: Four goals are expected to be achieved in the following experiments.

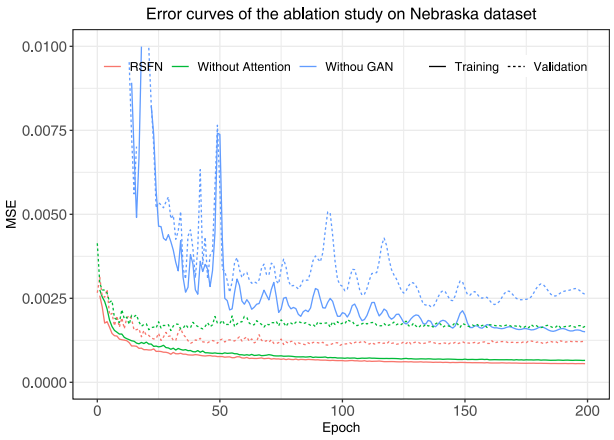


Fig. 4. Error curves of the ablation study in the Nebraska area.

1) To explore how the designed architecture affects the model effectiveness with the ablation study. Specifically, the key parts of the RSFN, namely, the GAN and attention, were removed using the Tianjin and Nebraska datasets with certain cloud contamination to understand their contribution to the overall model.
2) To investigate the fusion accuracy using the ideal datasets. In this case, the CIA and LGC datasets were used to demonstrate the fusion accuracy with cloud-free inputs instead of focusing on the fault tolerance.
3) To verify the fault tolerance capability of the model with noisy inputs. Both simulated and actual experiments were conducted to test the performance of the proposed RSFN model. Since the CIA and LGC datasets are free of clouds, simulated experiments were conducted by adding Gaussian noise to parts of the images randomly, and the corrupted references can be simulated to explore the fault tolerance of the models. The actual experiments conducted on Tianjin and Nebraska datasets aim to further test the fault tolerance capabilities of each model and compare the fusion accuracy with real data.
4) To study the timeliness of the references for fusion accuracy. This is the very reason why the RSFN model was proposed in the first place. Two test cases are designed to demonstrate the importance of timeliness by using the noisy references temporally close to the prediction date and using the ideal references far from the prediction date after quality filtering on the Tianjin and Nebraska datasets. The cloud-contaminated images were removed, and the remaining images were reorganized to obtain ideal inputs. Of course, the referential dates are relatively far from the prediction date in the cloud-free datasets. The noisy input case study still used the originally prepared datasets.

2) Models and Evaluation Metrics: Since the experiments mainly aim to evaluate the fault tolerance of STFMs with dual temporal references, models with two-reference inputs, including ESTARFM [11], StfNet [25], EDCSTFN [16], and RSFN, are selectively tested here. The ESTARFM belongs to the conventional model, while the other three are deep-learning-based models. The quantitative metrics used for the

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TABLE II

QUANTITATIVE METRICS ON THE CIA DATASET FOR EACH MODEL IN THE NEBRASKA STUDY AREA

	RMSE				SAM				SSIM				ERGAS			
	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN
20011008 20011017 20011102	0.023	**0.017**	0.019	**0.017**	0.259	0.176	0.177	**0.155**	0.964	0.982	0.978	**0.983**	53553	**43117**	50137	44556
20011017 20011102 20011109	0.028	**0.020**	0.024	0.021	0.308	0.183	0.213	**0.169**	0.944	**0.980**	0.967	0.978	62811	**48250**	60324	52326
20011102 20011109 20011125	**0.020**	0.029	0.023	0.024	0.219	0.335	**0.217**	0.252	**0.980**	0.945	0.959	0.951	**51703**	56393	57439	52061
20011109 20011125 20011204	**0.020**	0.028	0.022	0.024	0.206	0.298	**0.192**	0.236	**0.979**	0.947	0.959	0.950	45604	43748	43632	**40486**
20011125 20011204 20020105	0.029	**0.021**	0.025	0.022	0.289	0.177	0.194	**0.164**	0.942	**0.980**	0.970	**0.980**	49845	**41371**	48487	45524

TABLE III

QUANTITATIVE METRICS ON THE LGC DATASET FOR EACH MODEL IN THE NEBRASKA STUDY AREA

	RMSE				SAM				SSIM				ERGAS			
	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN
20041212 20041228 20050113	**0.019**	0.020	**0.019**	0.020	0.177	0.196	**0.165**	0.171	0.982	0.978	**0.983**	0.982	56759	61735	**53712**	56889
20041228 20050113 20050129	0.017	0.018	**0.015**	0.018	0.121	0.153	**0.111**	0.143	0.990	0.986	**0.991**	0.988	34575	35991	**31380**	35892
20050113 20050129 20050214	**0.018**	0.019	0.019	**0.018**	**0.125**	0.147	0.139	0.144	**0.989**	0.988	0.987	0.987	**35607**	37784	37288	36566
20050129 20050214 20050302	**0.013**	**0.013**	0.014	0.015	**0.090**	0.099	0.102	0.110	**0.994**	**0.994**	0.993	0.993	24338	**24203**	25759	29052
20050214 20050302 20050403	**0.012**	0.013	0.015	0.015	**0.092**	0.102	0.105	0.105	**0.995**	0.994	0.992	0.993	**20911**	23257	27028	25844

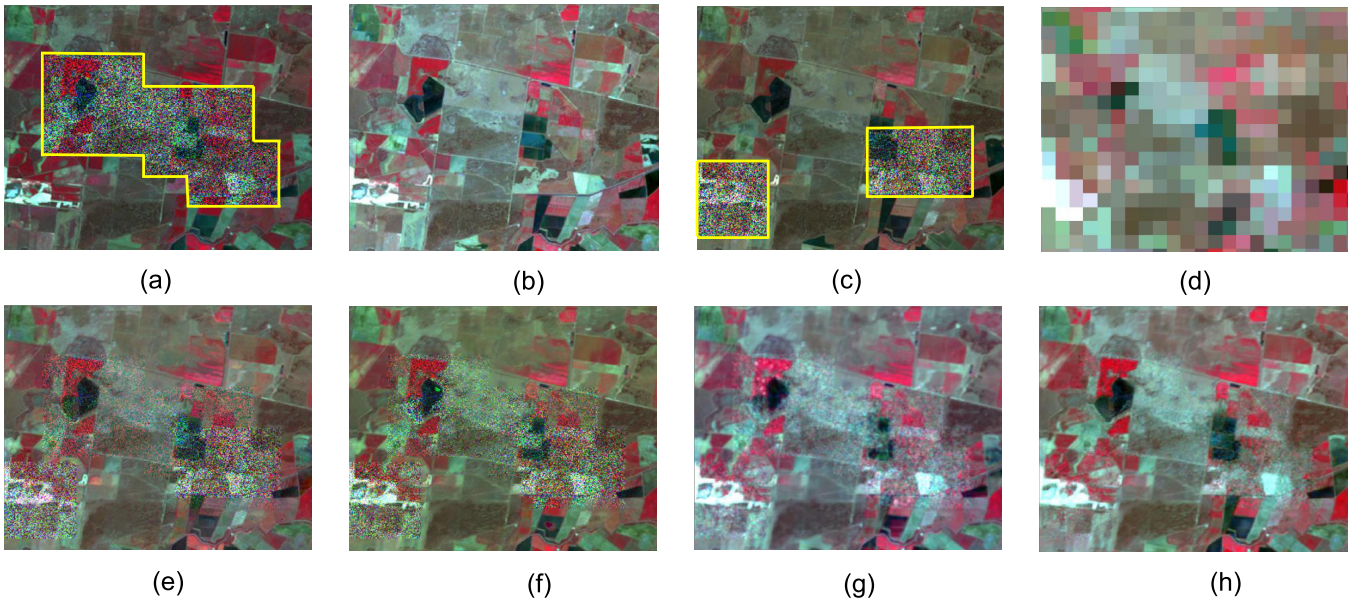


Fig. 5. Predicted results of each model on November 2, 2001, in part of the CIA study area. (The first row shows the three input images for the RSFN model and the ground truth on the prediction date. The Landsat references are contaminated with the simulated Gaussian noise, and the noisy sections are surrounded with yellow rectangles. Notably, the other three models need two additional MODIS references. The second row exhibits the predicted results of each model.) (a) Landsat (October 17, 2001). (b) GT (November 2, 2001). (c) Landsat (November 9, 2001). (d) MODIS (October 17, 2001). (e) ESTARFM prediction. (f) StfNet prediction. (g) EDCSTFN prediction. (h) RSFN prediction.

following evaluation include the root mse (RMSE) [50], spectral angle mapper (SAM) [51], SSIM index measure [52], and relative dimensionless global error (ERGAS) [53]. The RMSE measures the pixel-based errors, and a smaller value indicates higher accuracy. The SAM reflects the spectral distortion among all the bands of the predicted result, and a small value means higher accuracy. The SSIM quantifies the image similarity visually, and a higher score represents

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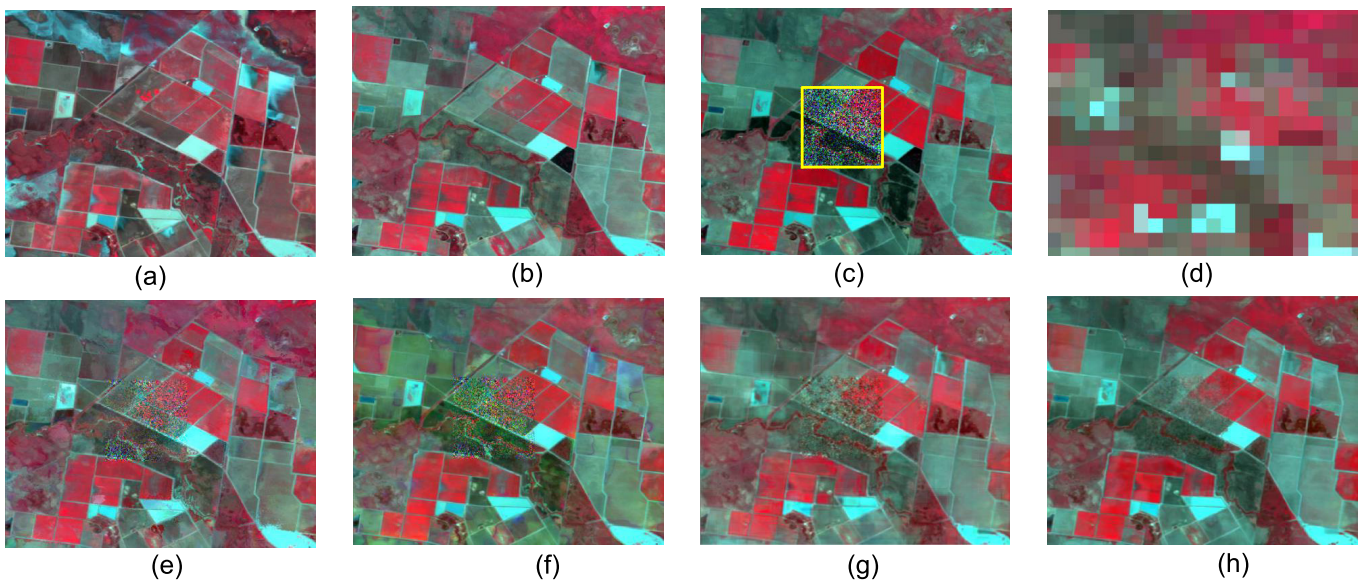


Fig. 6. Predicted results of each model on December 28, 2004, in part of the LGC study area. (The first row shows the three input images for the RSFN model and the ground truth on the prediction date. The Landsat references are contaminated with the simulated Gaussian noise, and the noisy sections are surrounded with yellow rectangles. Notably, the other three models need two additional MODIS references. The second row exhibits the predicted results of each model.) (a) Landsat (December 12, 2004). (b) GT (December 28, 2004). (c) Landsat (January 13, 2005). (d) MODIS (December 28, 2004). (e) ESTARFM prediction. (f) StfNet prediction. (g) EDCSTFN prediction. (h) RSFN prediction.

TABLE IV
QUANTITATIVE METRICS ON THE VALIDATION DATASET FOR EACH MODEL IN THE TIANJIN STUDY AREA

	RMSE				SAM				SSIM				ERGAS			
	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN
20190319 20190506 20191029	0.047	0.037	**0.032**	0.033	0.253	0.228	**0.195**	0.207	0.941	0.960	**0.968**	0.967	90461	77541	**66182**	70281
20200218 20200305 20200321	0.046	0.024	**0.019**	**0.019**	0.351	0.191	**0.152**	0.155	0.938	0.980	0.986	**0.987**	103820	50163	42058	**40401**
20200321 20200406 20200422	0.034	0.022	0.023	**0.022**	0.240	**0.149**	0.161	0.154	0.963	0.985	0.982	**0.984**	71643	41518	43266	**40864**
20200406 20200422 20200524	0.025	0.023	**0.021**	**0.021**	0.159	**0.145**	0.139	0.148	0.981	0.984	**0.985**	0.984	47424	43891	**41110**	41394
20200422 20200524 20200828	0.026	0.026	**0.023**	0.024	0.195	0.196	**0.178**	0.183	0.976	0.975	**0.980**	**0.980**	61129	60299	**55319**	56074
20200524 20200828 20201116	0.033	0.036	**0.027**	0.028	0.279	0.263	**0.230**	0.237	0.961	0.950	**0.970**	0.968	91087	99381	**81711**	83955

a better result. The ERGAS is an RMSE-based index tailored for spatiotemporal fusion, and a lower value shows high model performance.

3) Model Implementation and Settings: The deep fusion models were implemented using the PyTorch deep learning framework with Python programming language. The source code of this article is unveiled in the GitHub[1] platform for experimental reproduction and knowledge sharing. The models were trained on a machine equipped with 125-GB RAM and two NVIDIA RTX 3090 GPUs. The hyperparameters for the StfNet and EDCSTFN models followed the settings specified in the published papers [16], [25]. The numbers of the output channels for each block in the RSFN generator were empirically set to 16, 32, 64, 64, 32, 16, and 6. The last output channel equals the number of image bands to transform the feature maps to the original pixel space. The output channels of the hidden layers for the RSFN discriminator were empirically set to 32, 32, 64, 64, 128, 128, 256, 256, and 1. The input channel is twice the number of image bands because the conditional MODIS label was stacked with the Landsat-like prediction or the actual Landsat as the discriminator inputs. The total learnable parameters of each deep model are listed in Table I. The SftNet has the smallest model scale; the EDCSTFN and RSFN generator are comparable in model complexity. The Adam optimizer was employed to train the models, and the initial learning rate was set to 0.0005. The batch size for each epoch was set to 28 with a size of 256 × 256. The image slice operation is dynamically executed in

[1]https://github.com/theonegis/rsfn

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TABLE V
QUANTITATIVE METRICS ON THE VALIDATION DATASET FOR EACH MODEL IN THE NEBRASKA STUDY AREA

	RMSE				SAM				SSIM				ERGAS			
	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN	ESTARFM	StfNet	EDCSTFN	RSFN
20190713 20190729 20190814	0.025	**0.018**	**0.018**	**0.018**	0.399	0.243	0.203	**0.198**	0.966	0.984	0.986	**0.987**	57231	41898	42271	**40865**
20190814 20190915 20191017	0.032	0.027	**0.019**	0.020	0.473	0.278	**0.214**	0.215	0.953	0.967	**0.985**	0.983	79900	62864	**50393**	51132
20191017 20191102 20191118	0.026	**0.025**	0.026	0.026	0.186	0.181	0.169	**0.166**	0.979	0.980	**0.982**	0.980	39054	36191	**38033**	40688
20191102 20191118 20200222	0.030	0.026	0.023	**0.022**	0.202	0.152	0.133	**0.127**	0.974	0.983	0.986	**0.987**	44864	35679	32816	**31360**
20200222 20200613 20200731	0.086	0.075	0.052	**0.038**	0.558	0.415	0.235	**0.205**	0.842	0.886	0.946	**0.962**	114702	97039	66082	**52842**
20200731 20200816 20200917	0.061	0.032	0.025	**0.024**	0.693	0.282	0.244	**0.209**	0.883	0.954	0.976	**0.978**	140218	73358	61721	**59069**
20200816 20200917 20201206	0.039	0.029	0.027	**0.025**	0.341	0.250	0.216	**0.204**	0.939	0.965	0.972	**0.975**	82744	58832	50175	**48531**
20200917 20201206 20201222	**0.015**	0.017	0.028	0.021	**0.105**	0.106	0.159	0.112	**0.992**	**0.992**	0.981	0.990	**22900**	23346	38975	29424

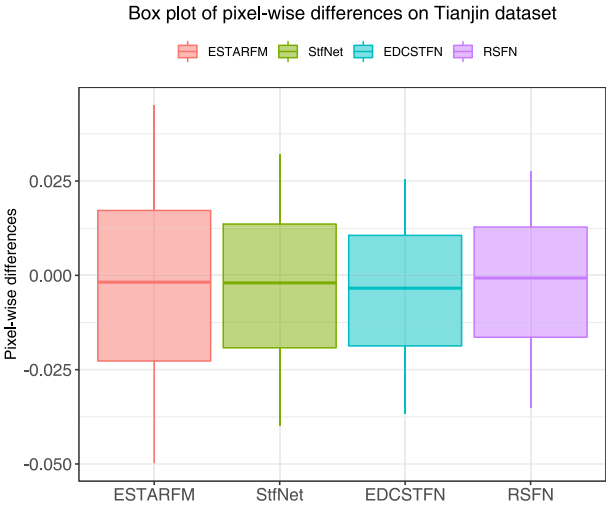


Fig. 7. Box plot of the pixelwise differences between prediction and target on validation data in the Tianjin area.

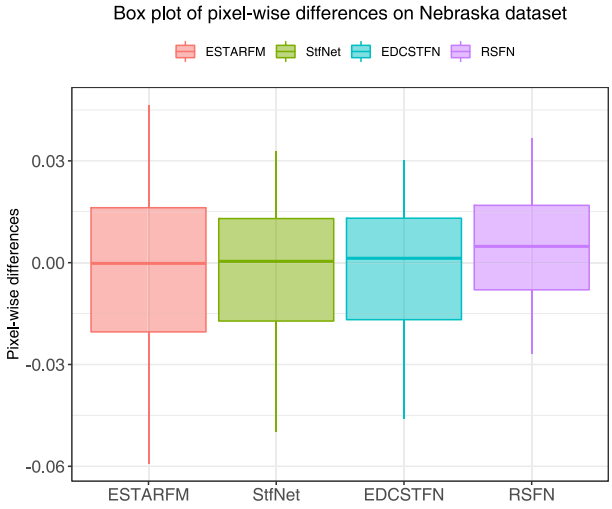


Fig. 8. Box plot of the pixelwise differences between prediction and target on validation data in the Nebraska area.

the training process. The training process was designed to iterate through the entire training dataset for 200 epochs in the experiments. All the deep models were trained in an end-to-end manner. The parameters of the loss function formulated in (1) were set as follows: $\alpha = \beta = \lambda = 1$. Of course, all these hyperparameters can be tweaked according to actual hardware specifications. Once the models are trained, the prediction can be completed within 1 s for all the models.

B. Ablation Study

Three architectures, including the original RSFN and the ones in which the attention subnet and GAN structure were removed, respectively, were tested, and the training curves are shown in Figs. 3 and 4. The prediction accuracy is indicated with mse. Several conclusions can be drawn from the above error curves.

1) The combination of GAN and attention can produce the highest accuracy in both study areas, which means that both the GAN and attention components can contribute to the model improvements.
2) The overall accuracy of the model with only GAN-based architecture is higher than the model with attention mechanism, which means that the GAN architecture affects the model performance more than the attention mechanism does.
3) The model convergence rate is slowest only with the plain CNN and attention mechanism. The differences with and without attention are obviously larger on the

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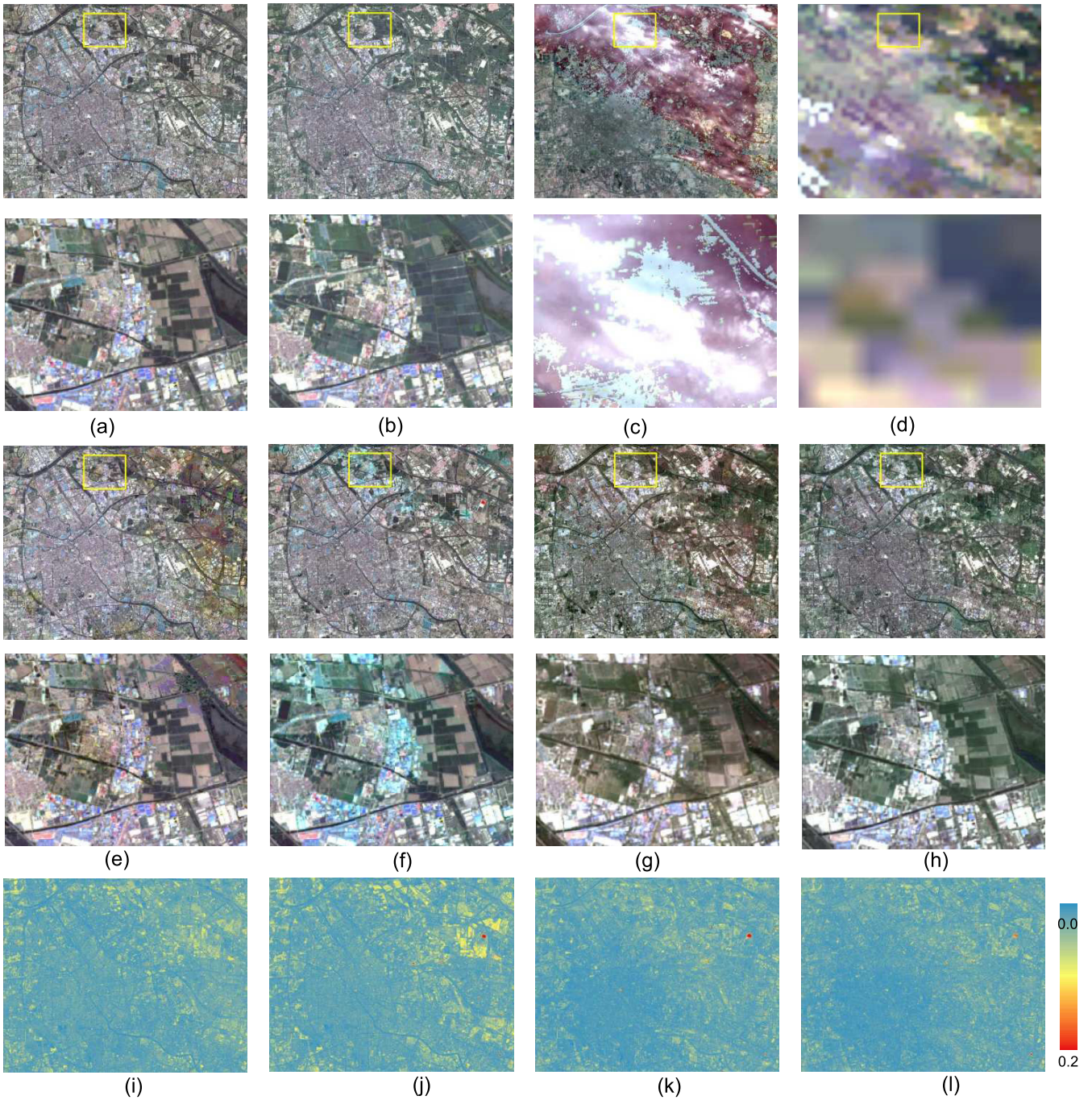


Fig. 9. Predicted results of each model on August 28, 2020, in part of the Tianjin area. (All the images are rendered with true color composition. The first row shows the three input images for the RSFN model and the ground truth on the prediction date. The other three models need two additional MODIS references. The zoomed-in details in the lower left corner correspond to the area marked with a yellow rectangle in the MODIS image. The second row exhibits the predicted results, and the last row shows the error maps by comparing the predictions with the ground truth.) (a) Landsat (May 24, 2020). (b) GT (August 28, 2020). (c) Landsat (November 16, 2020). (d) MODIS (August 28, 2020). (e) ESTARFM prediction. (f) StfNet prediction. (g) EDCSTFN prediction. (h) RSFN prediction. (i) ESTARFM errors. (j) StfNet errors. (k) EDCSTFN errors. (l) RSFN errors.

Nebraska dataset than that on the Tianjin dataset. This might be because large-scale thick clouds are present in some of the training data groups in Nebraska, and the attention mechanism is more effective in recognizing these distinct noises. In other words, the improvement from attention is more pronounced in the situation where considerable noise exists in the data.

To sum up, the GAN architecture contributes more to the fault-tolerant capability of the model, and the combination of GAN and attention can achieve the best performance. Therefore, both of them are necessary components in the RSFN model.

C. Test on the CIA and LGC Datasets

First, the fusion accuracy was tested on the ideal cloud-free datasets in CIA and LGC. The quantitative evaluations for each model on the validation data pairs are listed in Tables II and III. The RMSE was calculated based on the surface reference

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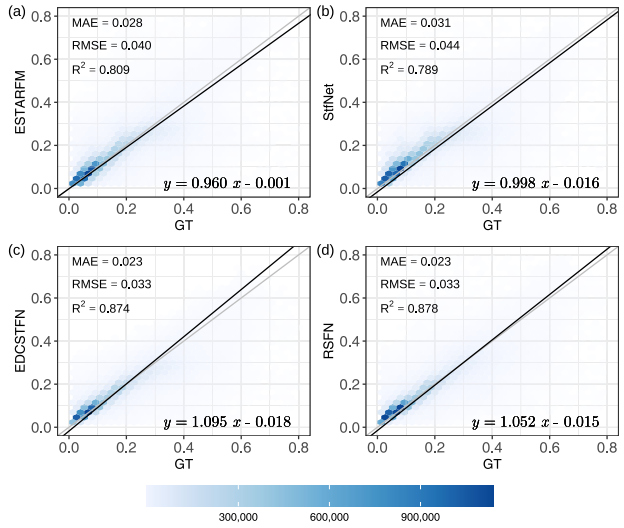


Fig. 10. Correlation between the prediction and ground truth on August 28, 2020, in Tianjin (The color of the small hexagon in the density plot denotes the number of points located in the corresponding location. The black line in the plot is the best fitting straight line using the least-squares regression, and the gray line is the identity line for comparison).

with a value range between 0 and 1, and the other three metrics were based on the pixel values with a scale factor of 10 000. All the metrics were calculated using an 8×8 sliding window for an accurate evaluation. The best of each metric is highlighted in bold. The first column of each row indicates the referential and prediction dates for the fusion cases. The dates are ordered in the sequence of the preceding date of the prediction, the prediction date, and the succeeding date of the prediction. Generally, no definitive conclusion can be drawn from the statistics on which model can achieve the highest fusion accuracy. Some performed better in this case, while some showed higher accuracy in that case. In other words, they stayed at almost the same level in the CIA and LGC datasets with cloud-free inputs.

Second, the fusion process was performed with simulated noisy inputs. The predicted results for part of the CIA and LGC study areas of each model on November 2, 2001, and December 28, 2004, are shown in Figs. 5 and 6, respectively. All the images are rendered with standard false-color composition to highlight the changes in the ground features. The first row exhibits the three input images for the RSFN model and the ground truth of Landsat in Figs. 5(b) and 6(b). All the Landsat references are contaminated with randomly distributed Gaussian noise in some regions. Visual observation from the second row shows that the predictions of the RSFN achieve the highest fault tolerance to the input noise and resemble the ground truth the most for both study areas. Especially, the noisy areas in the two referential images have an intersection in Fig. 5, though the RSFN model can still generate the noiseless prediction. In contrast, the predictions of the other three are all contaminated with different magnitudes of noise.

D. Test on the Tianjin and Nebraska Datasets

The experiments on fault tolerance capabilities were conducted using the Tianjin and Nebraska datasets with cloud contamination. The quantitative evaluations on the validation data pairs in the two areas are presented in Tables IV and V. Generally, the EDCSTFN and RSFN models perform at approximately the same level in Tianjin, while the RSFN model outperforms the others in most cases in Nebraska in terms of the quantitative metrics. This might be because the cloud contamination in some groups of Nebraska is more severe than that in Tianjin, and this somewhat suggests the superiority of the RSFN model with imperfect inputs. The statistics on the pixelwise differences for all validation data are illustrated with box plots in Figs. 7 and 8. The differences are calculated by subtracting the Landsat ground truth from predictions in a pixelwise manner. The prediction differences of ESTARFM and StfNet have a wider interquartile range on the Tianjin dataset, and the EDCSTFN and RSFN produce similar quartiles except that the median of the RSFN's differences is more closer to zero. The RSFN on the Nebraska dataset shows the most narrow interquartile range, but the medians of the other three models are more close to zero. A similar conclusion can be drawn that the proposed RSFN generally performs better than the ESTARFM and SftNet, and shows slight improvements over the EDCSTFN.

The predicted results in part of the Tianjin area of each model on August 28, 2020, are shown in Fig. 9. First, the referential image on November 16, 2020, contains evident noise in the upper right part of the image. Second, significant ground changes mainly fall into the suburb area in the upper right section, and the urban core area does not change much. Surprisingly, all the models can handle certain input noise although the other three models did not consider fault tolerance in the beginning. However, the prediction of the RSFN model in the suburban area is visually the closest to the ground truth. The density plots between the model predictions and Landsat ground truth are shown in Fig. 10. The evaluation metrics for the regression in the upper left corner suggest that the RSFN has the best fitness. Moreover, the black and gray lines in Fig. 10(d) of the RSFN model are the closest, showing the highest accuracy.

Fig. 11 exhibits the predicted results in part of the Nebraska area on June 13, 2020. The referential image on July 31, 2020, is severely contaminated with thick clouds, and there are significant ground changes because of the large timespan between the clean Landsat reference and the MODIS image on the prediction date. In this case, only the RSFN model can function properly and is barely affected by the input noise. The predictions of the ESTARFM and SftNet are severely contaminated with clouds. Although the EDCSTFN model performs better than the ESTARFM and SftNet models, the shapes and outlines of the clouds in the referential image are still present in the prediction of the EDCSTFN, which means that the EDCSTFN is not immune to the input data noise. The error maps of the last row in Fig. 11 and the density plots in Fig. 12 on June 13, 2020, also reconfirm the conclusion that the RSFN is fully capable of handling input noise and shows the highest robustness.

E. Test on the Timeliness of References

The qualitative experimental results for testing the timeliness of referential images are presented in Tables VI and VII.

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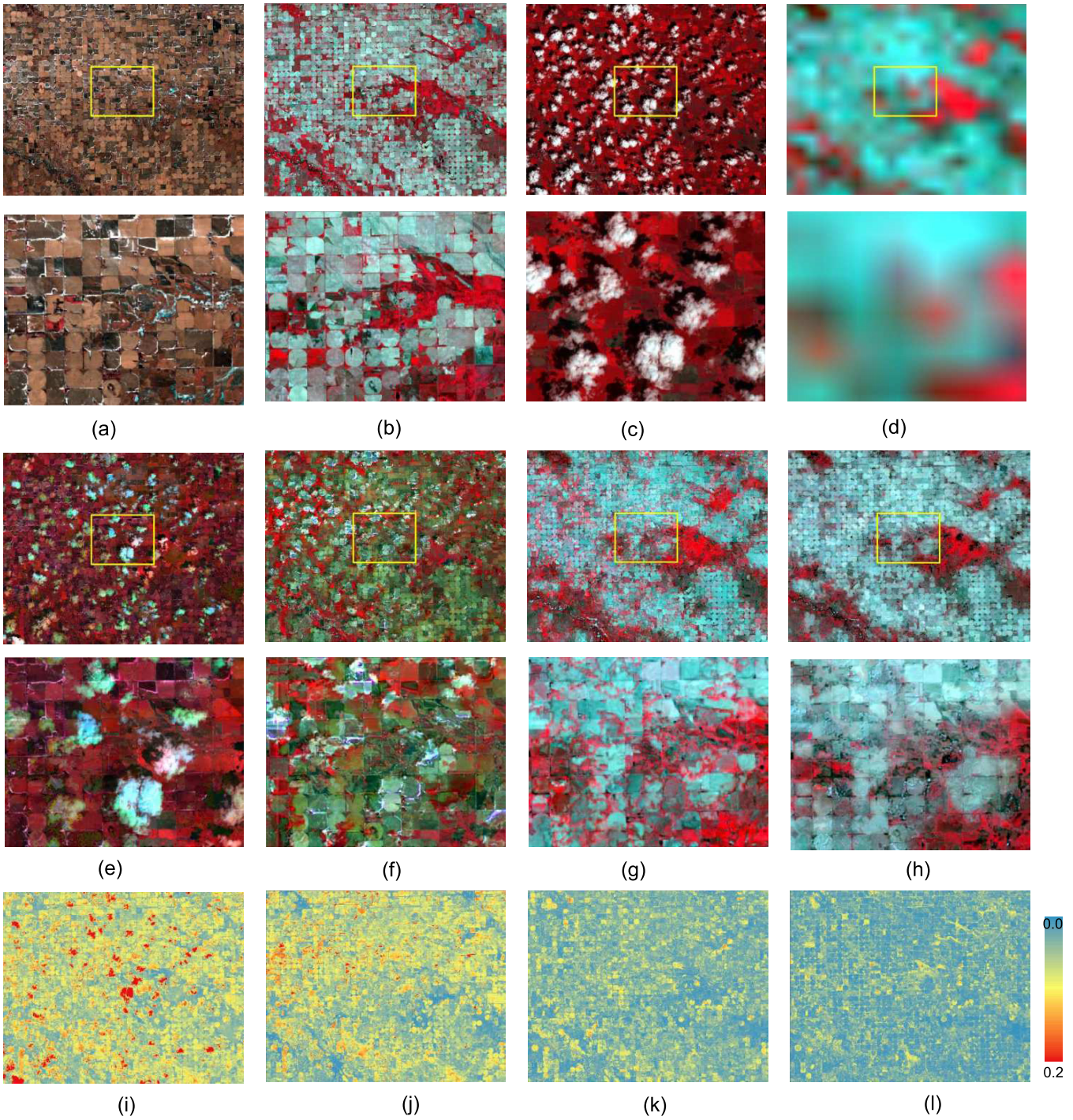


Fig. 11. Predicted results of each model on June 13, 2020, in part of the Nebraska area. (All the images are rendered with the standard false-color composition to highlight the crop changes. The first row shows the three input images for the RSFN model and the ground truth on the prediction date. The other three models need two additional MODIS references. The zoomed-in details in the lower-left corner correspond to the area marked with a yellow rectangle in the MODIS image. The second row exhibits the predicted results, and the last row shows the error maps by comparing the predictions with the ground truth.) (a) Landsat (February 22, 2020). (b) GT (June 13, 2020). (c) Landsat (July 31, 2020). (d) MODIS (June 13, 2020). (e) ESTARFM prediction. (f) StfNet prediction. (g) EDCSTFN prediction. (h) RSFN prediction. (i) ESTARFM errors. (j) StfNet errors. (k) EDCSTFN errors. (l) RSFN errors.

The model names with suffix "CF" indicate that fusion is conducted with cloud-free inputs. The RSFN model predictions with input noise are still listed here for comparison purposes. The two notations in the first column of each row represent the referential and prediction dates for fusion cases with input noise using the RSFN model and cases with cloud-free inputs after data filtering using the four models, respectively. Identical cases are omitted for brevity. Apparently, the timespans in the cases with the "CF" suffix are larger than those in the RSFN fusion cases with noise inputs. The qualitative metrics show that the RSFN model with a small timespan between the prediction and referential dates outperforms other models with

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TABLE VI

QUANTITATIVE METRICS ON THE VALIDATION DATASET FOR EACH MODEL IN THE TIANJIN STUDY AREA WITH CLOUD-FREE REFERENCES

	RMSE					SAM					SSIM					ERGAS				
	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN
20190319-20190506-20191029 20181026-20190506-20200305	0.046	0.041	0.036	0.038	**0.033**	0.247	0.233	0.213	0.227	**0.207**	0.942	0.955	0.961	0.959	**0.967**	88846	81416	72208	76874	**70281**
20200218-20200305-20200321 20190506-20200305-20200406	0.049	0.032	0.027	0.030	**0.019**	0.297	0.214	0.176	0.193	**0.155**	0.939	0.968	0.977	0.972	**0.987**	110594	71109	59822	63962	**40401**
20200321-20200406-20200422 20200305-20200406-20200422	0.041	0.023	0.024	**0.022**	**0.022**	0.257	0.162	0.166	0.164	**0.154**	0.950	0.981	0.977	0.981	**0.984**	86060	44494	47115	42943	**40864**
20200406-20200422-20200524 20200406-20200422-20201116	0.052	0.031	0.030	0.030	**0.021**	0.360	0.204	0.189	0.189	**0.148**	0.912	0.971	0.974	0.974	**0.984**	101168	64668	57174	57508	**41394**
20200422-20200524-20200828 20200422-20200524-20201116	0.034	0.035	0.029	0.030	**0.024**	0.257	0.248	0.218	0.231	**0.183**	0.962	0.957	0.971	0.969	**0.980**	81513	90120	70658	73575	**56074**
20200524-20200828-20201116 20200422-20200828-20201116	0.042	0.050	0.029	0.033	**0.028**	0.384	0.328	0.248	0.268	**0.237**	0.937	0.908	0.965	0.956	**0.968**	123119	146476	89977	99853	**83955**

TABLE VII

QUANTITATIVE METRICS ON THE VALIDATION DATASET FOR EACH MODEL IN THE NEBRASKA STUDY AREA WITH CLOUD-FREE REFERENCES

	RMSE					SAM					SSIM					ERGAS				
	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN	ESTARFM-CF	StfNet-CF	EDCSTFN-CF	RSFN-CF	RSFN
20190713-20190729-20190814 20190408-20190729-20190915	0.031	0.032	0.024	0.024	**0.018**	0.320	0.279	0.244	0.247	**0.198**	0.961	0.967	0.979	0.977	**0.987**	76046	91212	56586	57580	**40865**
20190814-20190915-20191017 20190729-20190915-20191102	0.025	0.025	**0.020**	**0.020**	**0.020**	0.253	0.248	0.220	**0.209**	0.215	0.975	0.974	**0.984**	0.983	0.983	66971	63374	56950	53383	**51132**
20191017-20191102-20191118 20190915-20191102-20191118	0.026	**0.025**	0.029	0.028	0.026	0.179	**0.162**	0.184	0.170	0.166	0.979	**0.982**	0.978	0.980	0.980	**40329**	40911	42951	43622	40688
20191102-20191118-20200222 20191102-20191118-20200613	0.031	0.030	0.026	0.029	**0.022**	0.164	0.165	0.137	0.143	**0.127**	0.978	0.977	0.984	0.982	**0.987**	45593	44081	36720	41881	**31360**
20200222-20200613-20200731 20191118-20200613-20200816	0.064	0.053	0.040	**0.036**	0.038	0.269	0.256	**0.173**	0.190	0.205	0.919	0.943	0.970	**0.968**	0.962	83697	70646	56040	**50707**	52842
20200731-20200816-20200917 20200613-20200816-20200917	0.053	0.058	0.033	0.039	**0.024**	0.635	0.391	0.293	0.330	**0.209**	0.901	0.878	0.959	0.940	**0.978**	118825	129293	78615	88884	**59069**

cloud-free inputs in nearly all cases, which shows the importance of the timeliness of reference data in spatiotemporal fusion. Figs. 13 and 14 exhibit the partial prediction results for the Tianjin area on August 28, 2020, and for the Nebraska area on June 13, 2020, respectively. Evidently, the predicted spectra of the RSFN model are closest to the ground truth, and the errors are the smallest among other cases.

IV. DISCUSSION

The classical weight-function-based method ESTARFM and the deep-learning-based SftNet were designed to perform fusion at the pixel level. If data repair is not performed in advance, the predicted results are easily affected by the input noise even though dual temporal references are employed. This can be clearly observed from the simulated experiments using the CIA and LGC datasets because, where there is noise contamination in the input images, there are defective pixels in the predicted images with the pixel-level fusion models, as shown in Figs. 5 and 6. Therefore, fusion models at the pixel level are incapable of handling input data noise without extra data preprocessing procedures.

In contrast, the fusion process conducted at the feature level in the EDCSTFN and RSFN becomes more insensitive to the input noise. The EDCSTFN learns the residuals between the referential and prediction and accomplishes fusion by adding the residual to the fine-resolution referential feature maps. The weights for the two references are based on the learned residuals. Sometimes, for inputs with limited noise, the EDCSTFN model might handle them well, as shown in Fig. 9. However, because the EDCSTFN model has no built-in mechanism to filter out input noise, in the cases with severe cloud contamination, the results would be corrupted, as shown in Fig. 11. Even though the EDCSTFN and RSFN can achieve similar performance regarding the statistical metrics in some cases, as shown in Table IV, the EDCSTFN cannot work properly when there are severe thick clouds in the referential images.

The RSFN, first, can automatically learn the weights using the attention subnet and weight the referential feature maps.

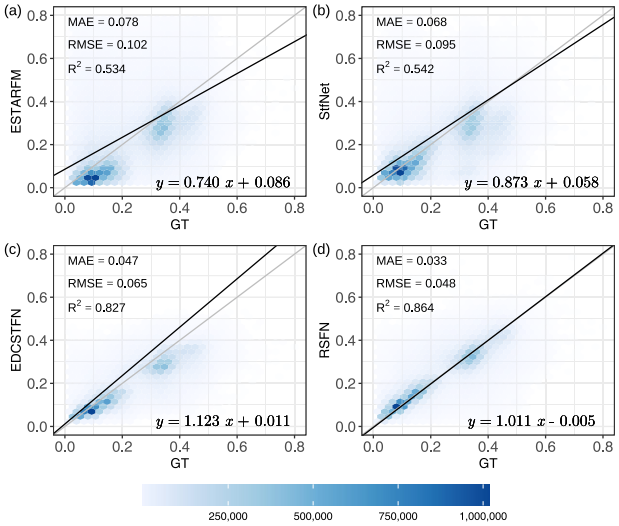


Fig. 12. Correlation between the prediction and ground truth on June 13, 2020, in Nebraska (The color of the small hexagon in the density plot denotes the number of points located in the corresponding location. The black line in the plot is the best fitting straight line using the least-squares regression, and the gray line is the identity line for comparison).

It is expected yield weights close to zero for the areas with severe input noise. Second, the inputs are curtailed to just three images, namely, the two Landsat references and one MODIS image on the prediction date. Because input data might be contaminated with noise, the reduction of the inputs means decreasing the possibility of introducing extra noise. Thus, the RSFN model only requires the input MODIS to be free of clouds and in good quality. Third, the GAN architecture is effective for image inpainting and restoration [54]; hence, the GAN-based RSFN naturally has a certain ability to restore an uncorrupted image to some extent. These three aspects might explain why the RSFN can gain higher robustness and fault tolerance than other models.

The experiments using the Tianjin and Nebraska datasets with cloud-free and noisy references show the importance of the timeliness of referential data. After data filtering, the cases with cloud-free inputs have a large timespan between

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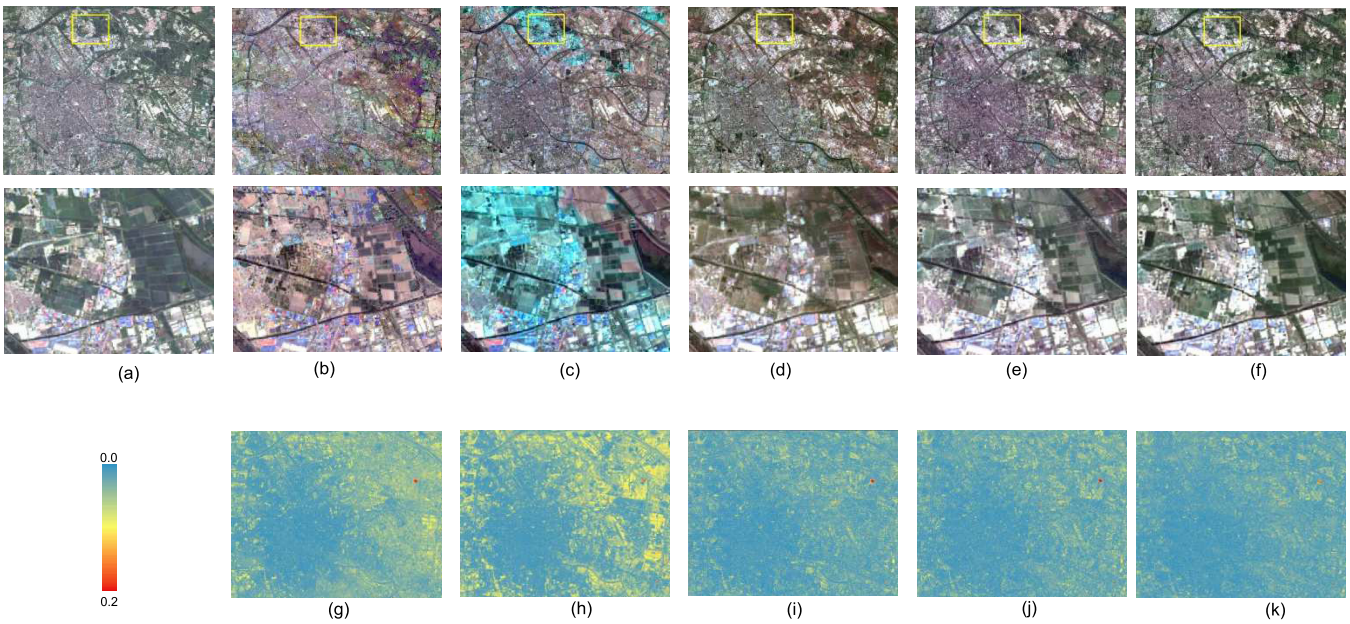


Fig. 13. Predicted results of each model on August 28, 2020, in part of Tianjin area. (The first row shows the Landsat ground truth and model predictions. The model name with the suffix "CF" indicates that the fusion is conducted with cloud-free inputs. The zoomed-in detail in the second row corresponds to the area marked with a yellow rectangle in the first row. The last row shows the error maps.) (a) GT (August 28, 2020). (b) ESTARFM-CF prediction. (c) StfNet-CF prediction. (d) EDCSTFN-CF prediction. (e) DRSTFN-CF prediction. (f) RSFN prediction. (g) ESTARFM-CF errors. (h) StfNet-CF errors. (i) EDCSTFN-CF errors. (j) DRSTFN-CF errors. (k) RSFN errors.

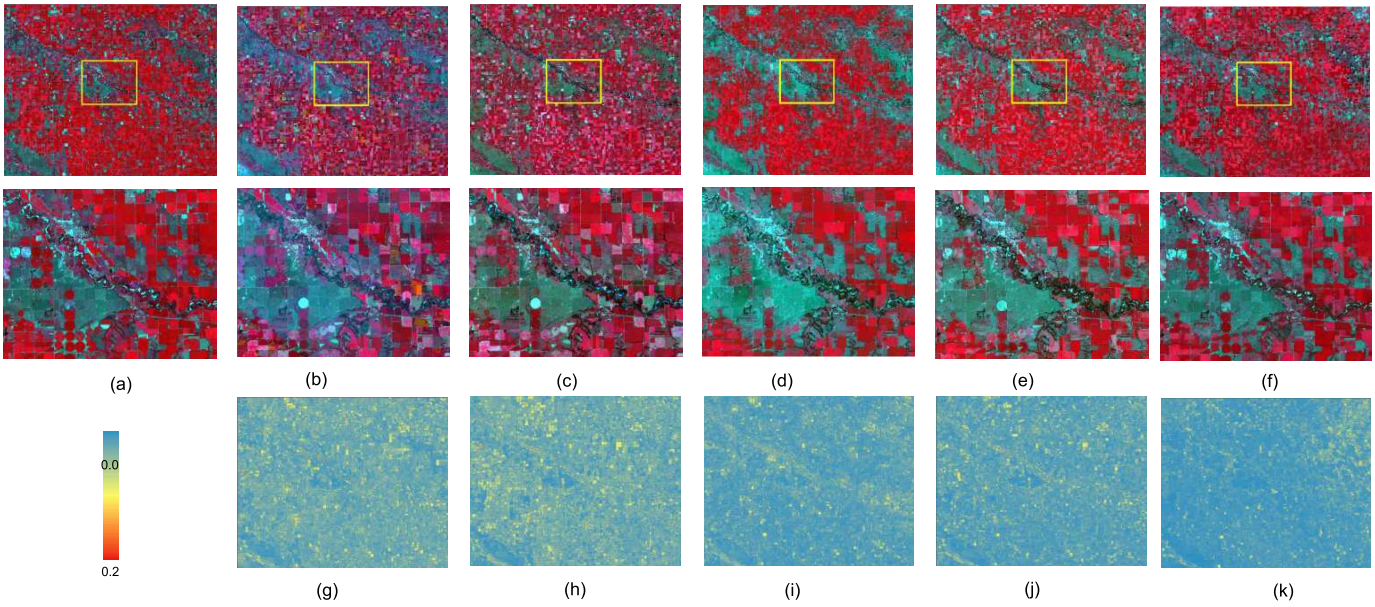


Fig. 14. Predicted results of each model on June 13, 2020, in part of Nebraska area. (The first row shows the Landsat ground truth and model predictions. The model name with the suffix "CF" indicates that the fusion is conducted with cloud-free inputs. The zoomed-in detail in the second row corresponds to the area marked with a yellow rectangle in the first row. The last row shows the error maps.) (a) GT (July 29, 2019). (b) ESTARFM-CF prediction. (c) StfNet-CF prediction. (d) EDCSTFN-CF prediction. (e) DRSTFN-CF prediction. (f) RSFN prediction. (g) ESTARFM-CF errors. (h) StfNet-CF errors. (i) EDCSTFN-CF errors. (j) DRSTFN-CF errors. (k) RSFN errors.

the referential and prediction dates, while the cases with a certain amount of noise can gain a small timespan between the referential and prediction dates. Theoretically, for a large timespan, the ground features may have changed significantly; however, the cases with a smaller timespan can preserve the ground surface information as much as possible. Thus, ground features can provide some referential information quite close to the prediction date temporally and further contribute to the fusion accuracy. Both the statistical metrics in Tables VI and VII and visual observation in Fig. 13 and 14 confirm the conclusion that the model with references temporally close to the prediction date outperforms the model with a large timespan even though the model with smaller-timespan inputs is contaminated with noise.

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In summary, fusion models at the pixel level cannot handle input noise properly, and fusion models at the feature level have a certain ability to automatically suppress input noise. With a smaller timespan between referential and prediction dates, even though the inputs are contaminated with a certain amount of noise, the RSFN model can predict fusion results unaffected with high robustness. Thus, we can conclude that the application scope of the STFM can be expanded to cloud-prone regions with the RSFN model, showing promising prospects in practical applications.

V. Conclusion and Prospects

Over the past few decades, significant progress has been made in the spatiotemporal fusion domain; however, most of the models require input data free of clouds and without missing data. Actual remote sensing data may not satisfy these requirements in practical applications. After filtering the dataset with quality control, the referential and prediction timespan is often enlarged, and distinct ground changes may occur; thus, the prediction accuracy may become lower accordingly. This article proposes an end-to-end STFM for remote sensing images named RSFN to automatically handle input data noise. By comparing with the other three fusion models, the experiments in four study areas demonstrate the robustness and fault tolerance capability of the proposed RSFN. Theoretically, the RSFN improves the model's robustness in three ways. First, the decreased number of inputs can partially reduce the possibility of introducing additional errors. Second, the input Landsat images with certain noise can be automatically corrected with feature-level weighted fusion by utilizing an attention-based subnet. Third, the GAN-based architecture possesses a certain ability to fix the uncorrupted image to some extent.

The experience of using different STFMs in practice shows that some models may perform excellently in some geographic areas on some datasets but fail to make accurate predictions in other areas on other datasets. After years of development, further boosting the model accuracy becomes increasingly difficult. In this situation, improving the model robustness and reliability for practical uses and applications requires urgent attention. The RSFN model is a preliminary attempt toward this direction. As far as we are considered, to further improve the model usability, we can proceed from the following three aspects. First, some issues regarding the model robustness need further exploration. For example, experiments should be conducted to qualify the maximum data input noise for a fault-tolerant model, or the maximum timespan between referential and prediction should have a referential standard under different circumstances. Second, the reliability of the predicted results should be qualified at the pixel level. Remote sensing images distributed by authorities usually ship with quality metadata, which is especially important for quantitative applications. The spatiotemporal fusion results should also provide quality metadata to describe the prediction reliability. Third, model transferability in different geographic areas or on different datasets should be explored. For example, if one fusion model is trained with some datasets, how can it be transferred to other datasets in other areas with minimum effort? These questions remain open to advanced study.

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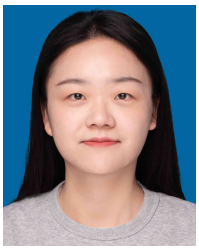
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